

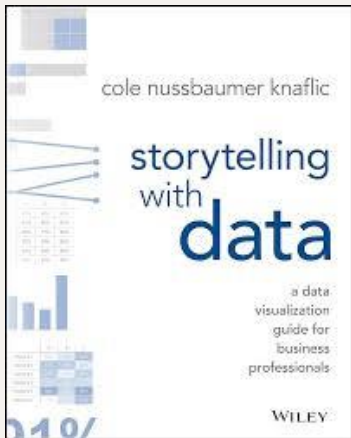
Data Visualization and Storytelling

Professor Laura Ruiz



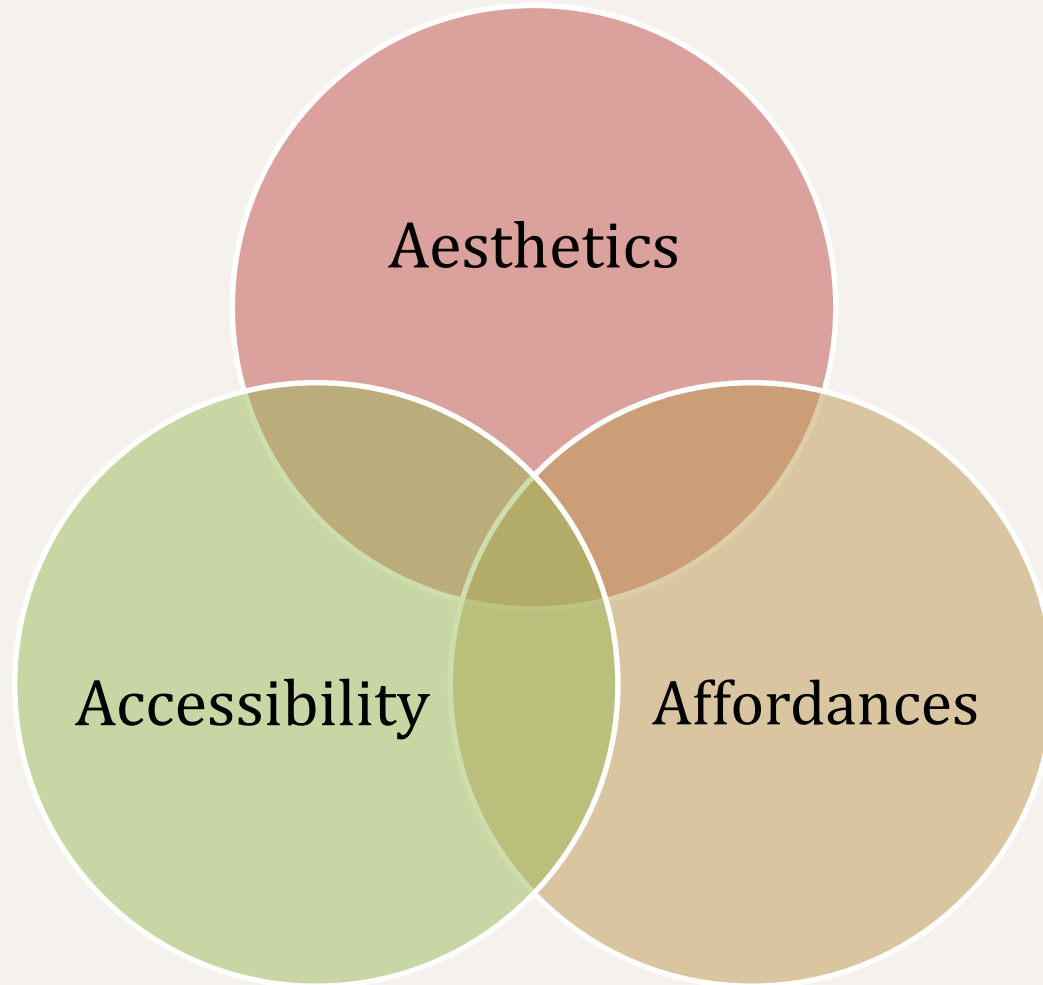
March 25, 2026

SESSION 6: Think like a designer, design sophistication, dashboard design



*Excellence in statistical graphics consists in
communicating complex ideas with clarity,
precision and efficiency.*

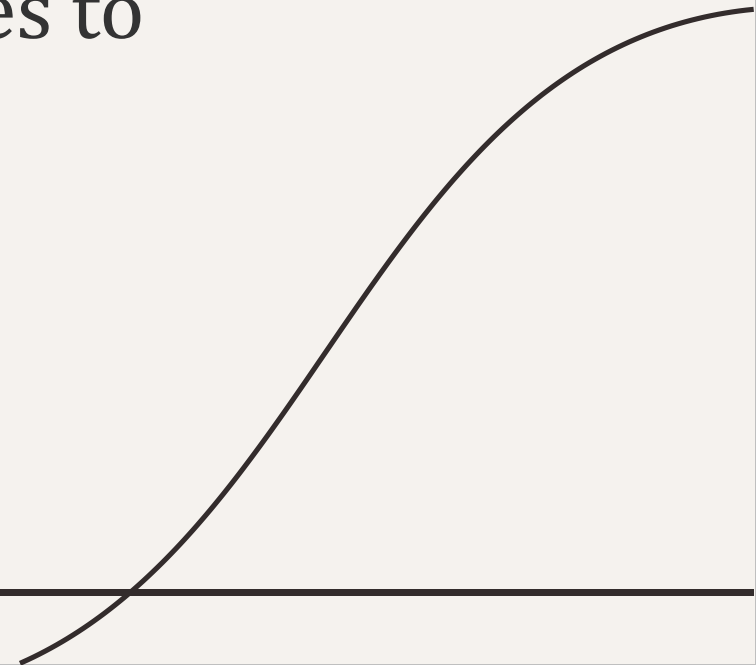
Thinking Like a Designer – Key Concepts



Affordances

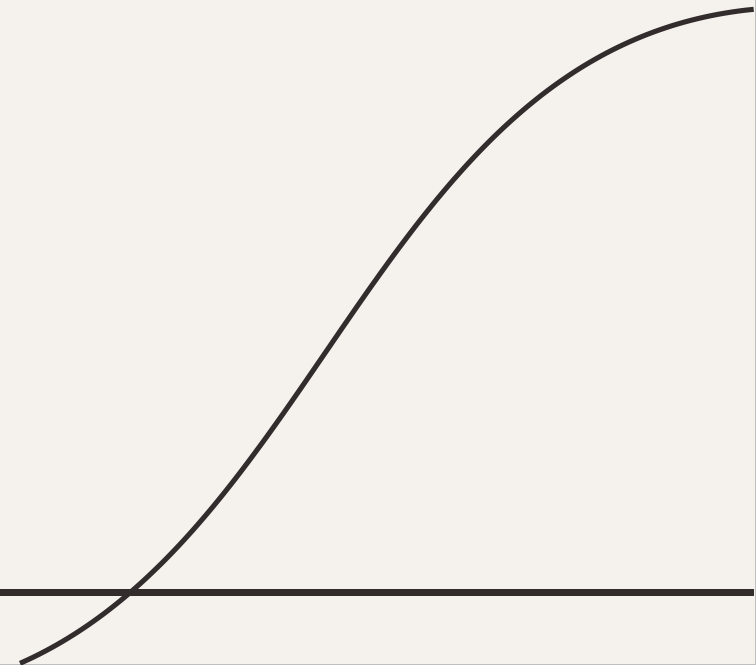
Affordances are aspects *inherent* to a design that make it obvious how the object is to be interacted with.

We have 3 major items to review when it comes to affordances



Affordances

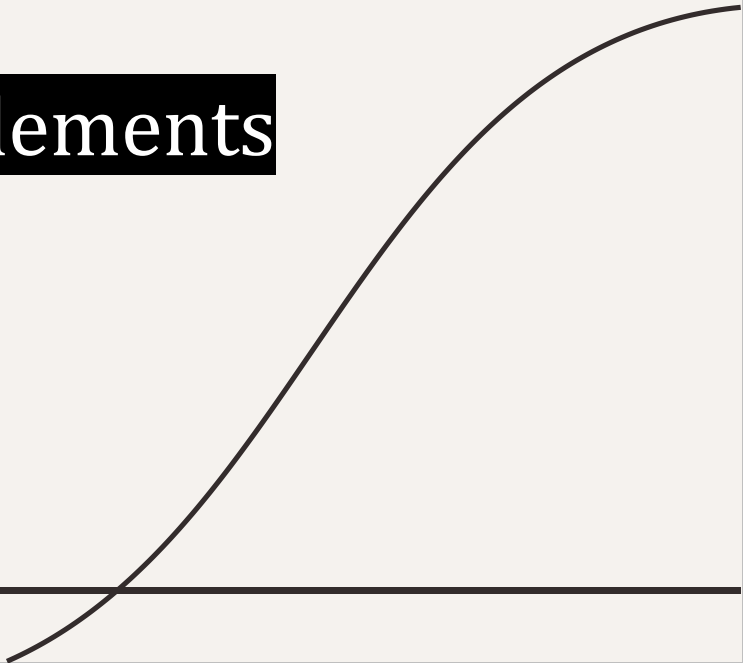
1. Highlight the important stuff
- ~~2. Eliminate distractions~~
3. Create a clear
 - hierarchy
 - of
 1. information



Affordances - Highlight

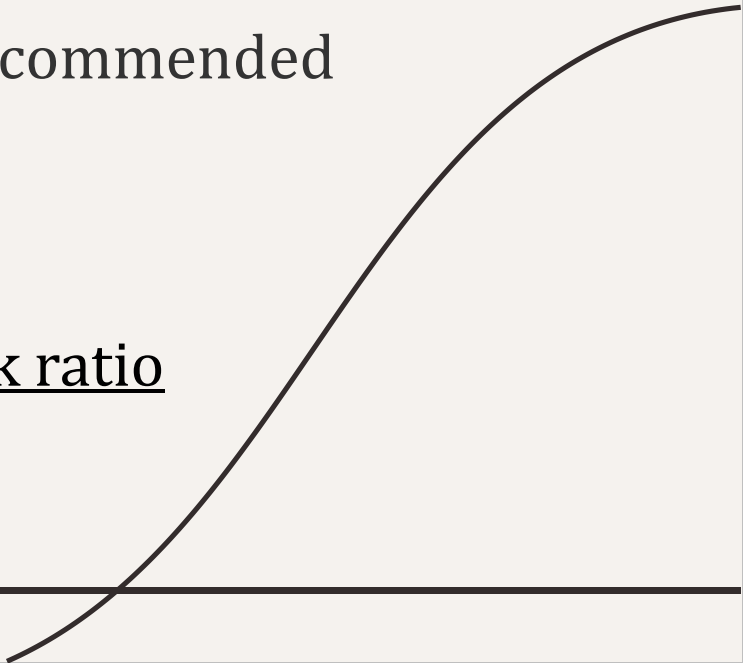
Remember pre-attentive attributes?

We can highlight information we want our audience to focus on – however, do so sparingly (about 10%)

- **Bold**, *Italics*, underlining
 - CASE and *typeface*
 - Color
 - **Inverse Elements**
 - Size
- 

Highlight the important stuff

In *Universal Principles of Design* (Lidwell, Holden, and Butler, 2003), it is recommended that at most 10% of the visual design be highlighted.

- **Bold, Italics, underlining**
 - Use for title, labels, captions, and short word sequences
 - CASE and *typeface*
 - Uppercase text in short words; typeface isn't usually recommended
 - **Color**
 - Effective, but again used sparingly
 - **Inverse Elements**
 - Effective at grabbing attention, but think of the data-ink ratio
 - **Size**
 - Effective to attract attention and signal importance
- 

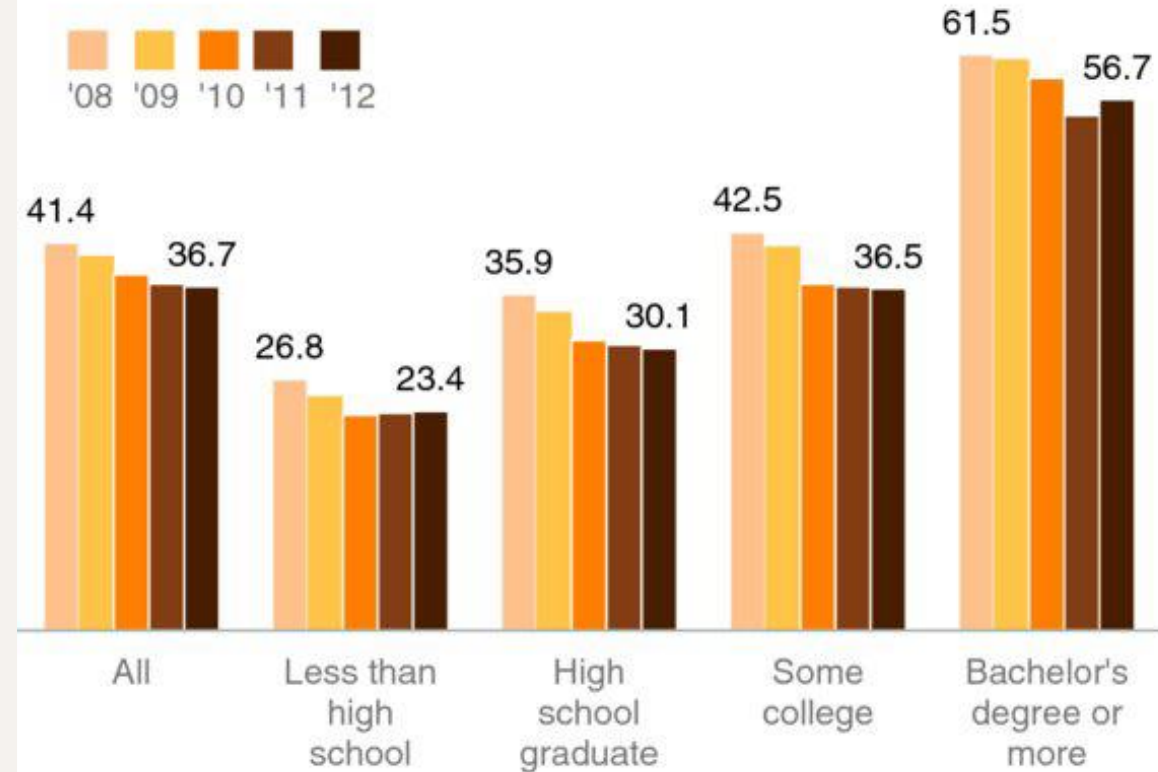
Example: Pew Research Center original graph

2011 to 2012 increase observed in total new marriages was driven primarily by an increase in those having a bachelor's degree or more.

Changing the use of color in this visual can completely redirect our focus.

New Marriage Rate by Education

Number of newly married adults per 1,000 marriage eligible adults



Note: Marriage eligible includes the newly married plus those widowed, divorced, or never married at interview.

Source: U.S. Census

Adapted from PEW RESEARCH CENTER

Eliminate distractions

“You know you’ve achieved perfection, not when you have nothing more to add, but when you have nothing to take away” (Saint-Exupery, 1943).

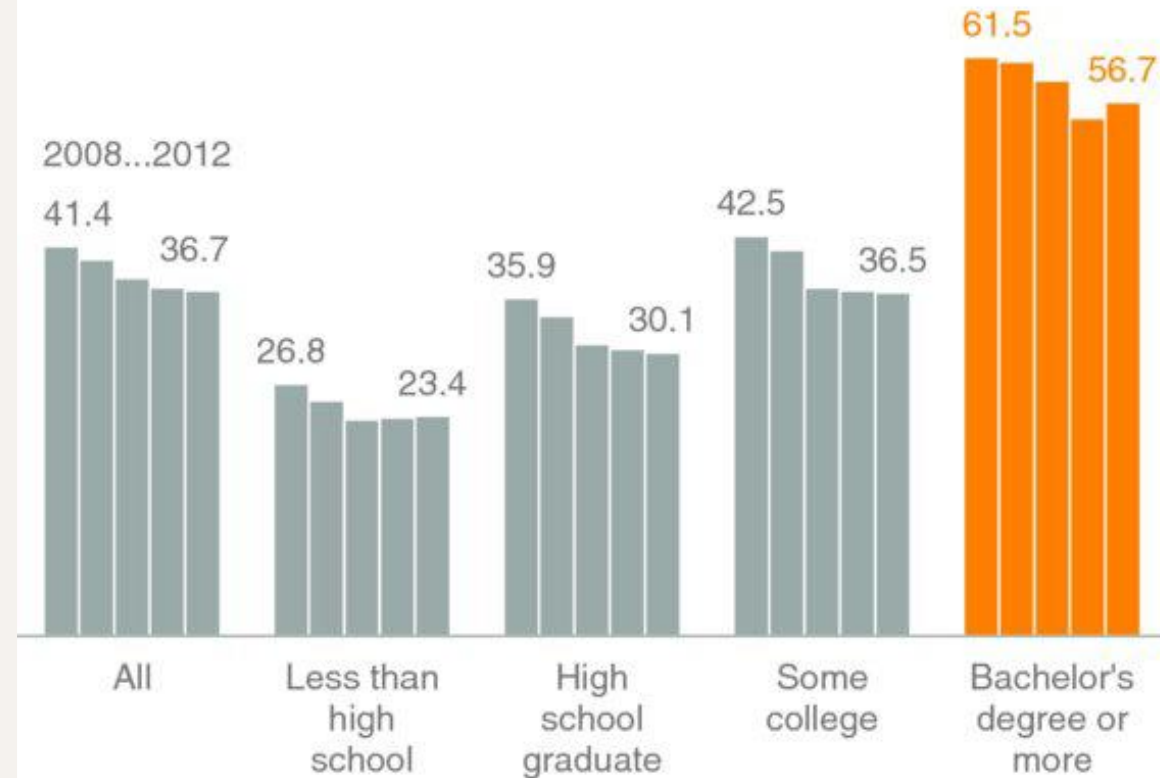
1. **Not all data are equally important.** Use your space and audience’s attention wisely by getting rid of noncritical data or components.
 2. **When detail isn’t needed, summarize.** You should be familiar with the detail, but that doesn’t mean your audience needs to be. **Consider whether summarizing is appropriate.**
 3. **Ask yourself: would eliminating this change anything?** No? Take it out! Resist the temptation to keep things because they are cute or because you worked hard to create them; if they don’t support the message, they don’t serve the purpose of communication.
 4. **Push necessary, but non-message-impacting items to the background.** Use your knowledge of preattentive attributes to de-emphasize. Light grey works well for this.
- 

Example: Pew Research Center original graph

How can we eliminate distractions
here?

New Marriage Rate by Education

Number of newly married adults per 1,000 marriage eligible adults



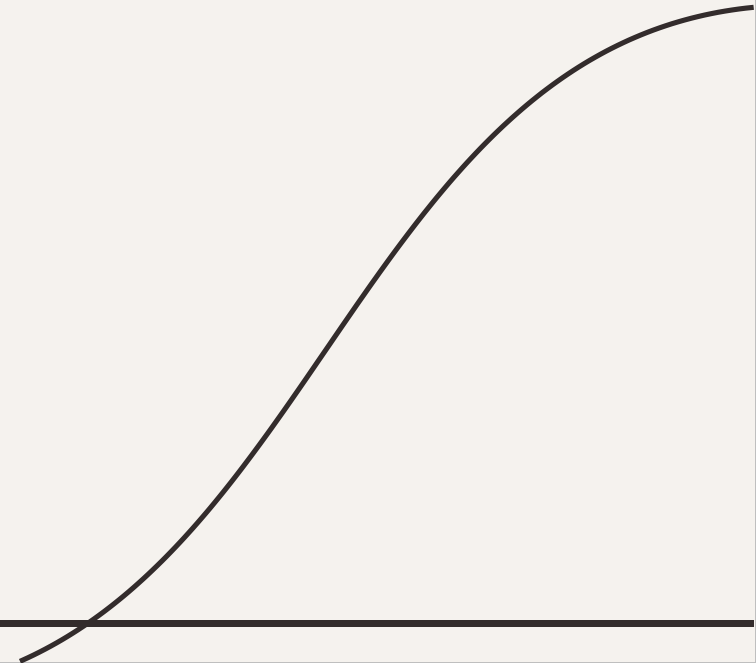
Note: Marriage eligible includes the newly married plus those widowed, divorced, or never married at interview.

Source: U.S. Census

Adapted from PEW RESEARCH CENTER

Affordances – **Visual Hierarchy**

Visual Hierarchy establishes the **order** for the audience to use to process the information without it feeling complicated



Afford

**You will
read this
first.**

AND YOU'LL
MOST LIKELY
LEAVE THIS
FOR LAST.

And then you'll probably read
this immediately afterwards.

Create a clear **visual hierarchy** of information

We can visually pull some items to the forefront and push other elements to the background, indicating to our audience the **general order** in which they should process the information we are communicating.

Imagine you are a car manufacturer. Two important dimensions by which you judge the success of a particular make and model are (1) customer satisfaction and (2) frequency of car issues.

Issues vs. Satisfaction by Model



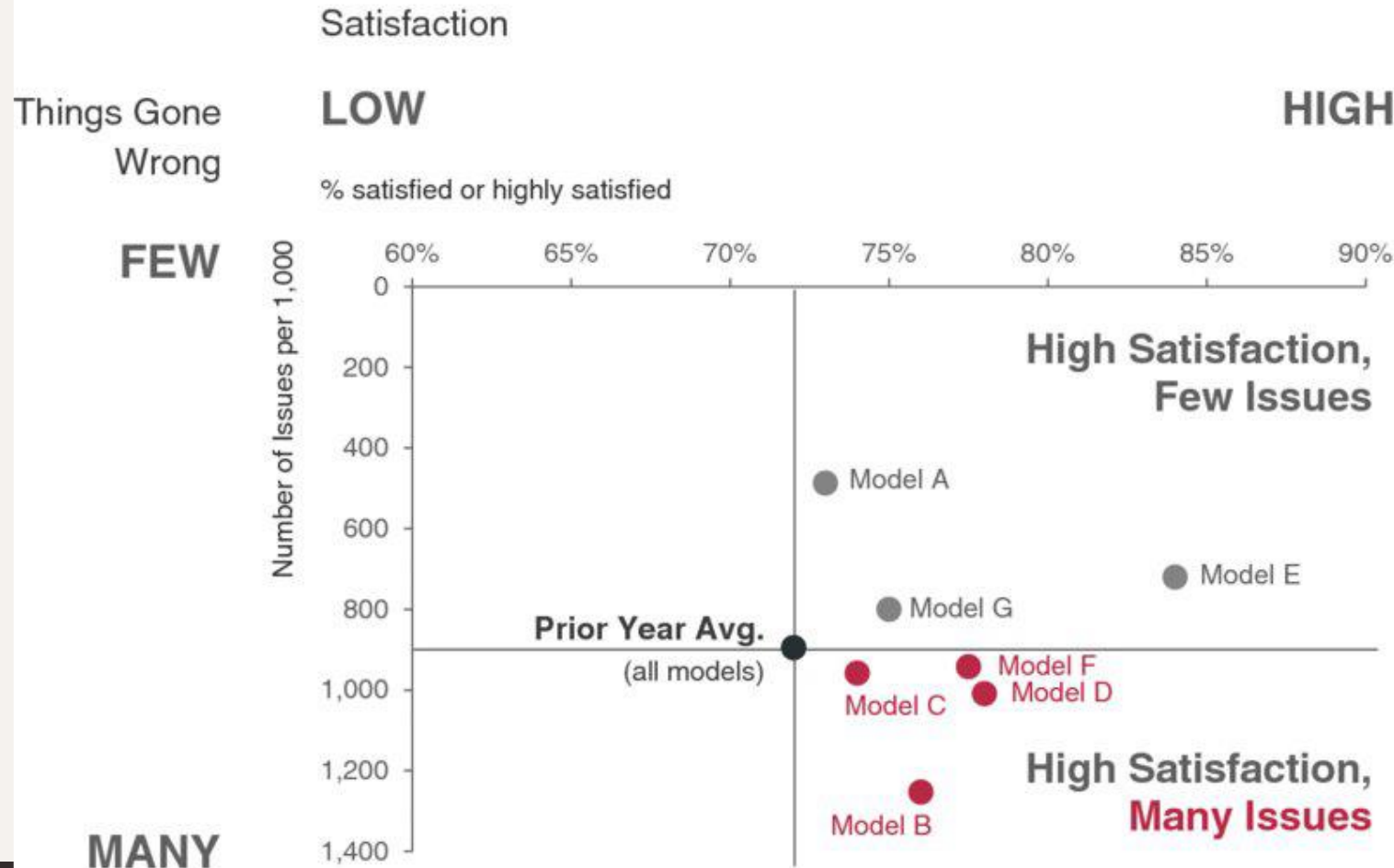
Create a clear **visual hierarchy** of information

The size and color of font and data points alert us where to pay attention and in what general order.

Red in the bottom right quadrant. The words tell me satisfaction is high, but there are many issues. It's clear because of how the visual is constructed that these are cases where the level of issues is greater than it was for last year's average.

The red color reinforces that this is a problem.

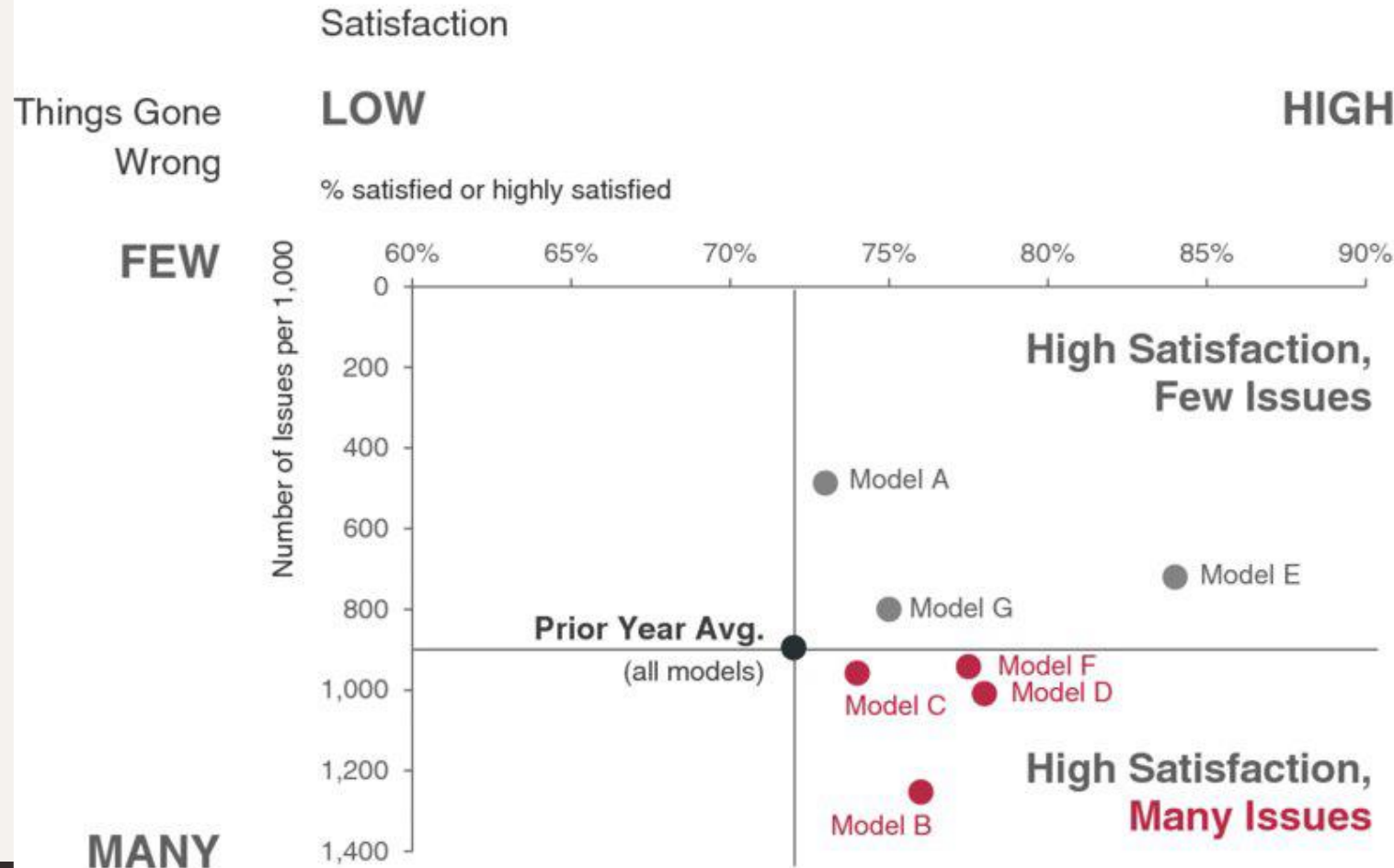
Issues vs. Satisfaction by Model



Create a clear **visual hierarchy** of information

Note that the left quadrants aren't labeled; labels are unnecessary since no values fall there.

Issues vs. Satisfaction by Model

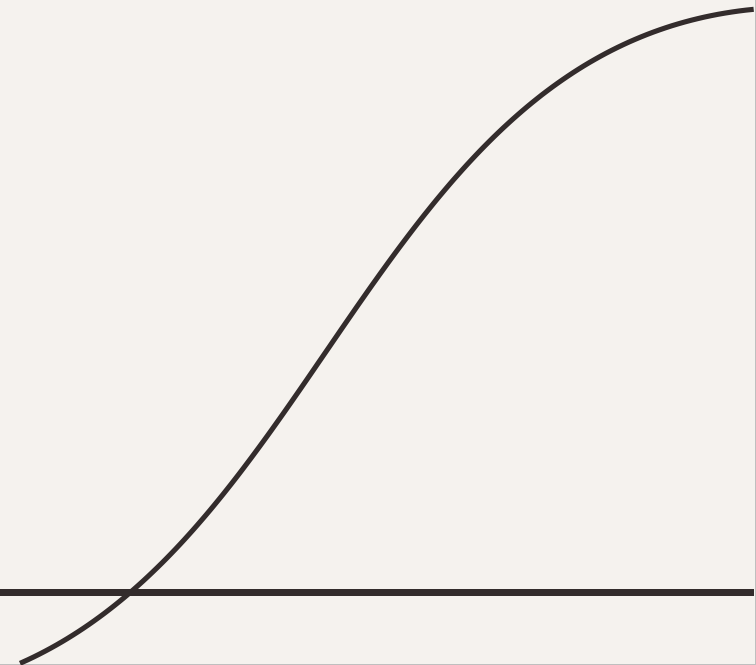


Accessibility

Generally, accessibility refers to designs that are usable by people of diverse abilities.

You might be an engineer , but you shouldn't need to get a master in engineering to understand what's going on in your visualization.

1. Don't overcomplicate
2. Text is your friend!

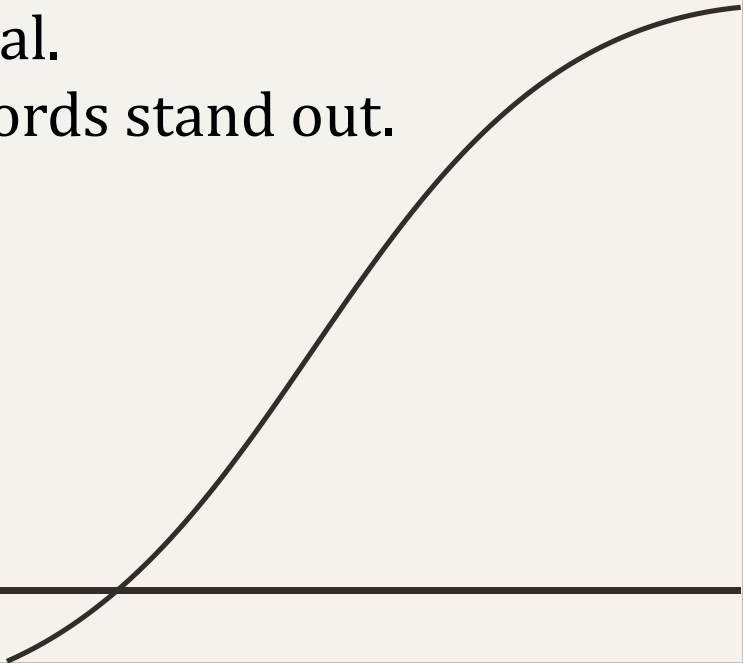


Accessibility – Don't overcomplicate

- Make it legible
 - Use consistent, easy to read font (no
◆◆)(■)∩⊂⊃(■)∩◆)
- Keep it clean
 - Use appropriate visual affordances
- Use straightforward language
 - Choose simple language; fewer words is generally better than many; define words; spell out acronyms
- Remove unnecessary complexity
 - KISS – keep it simple stupid!



Text is your friend

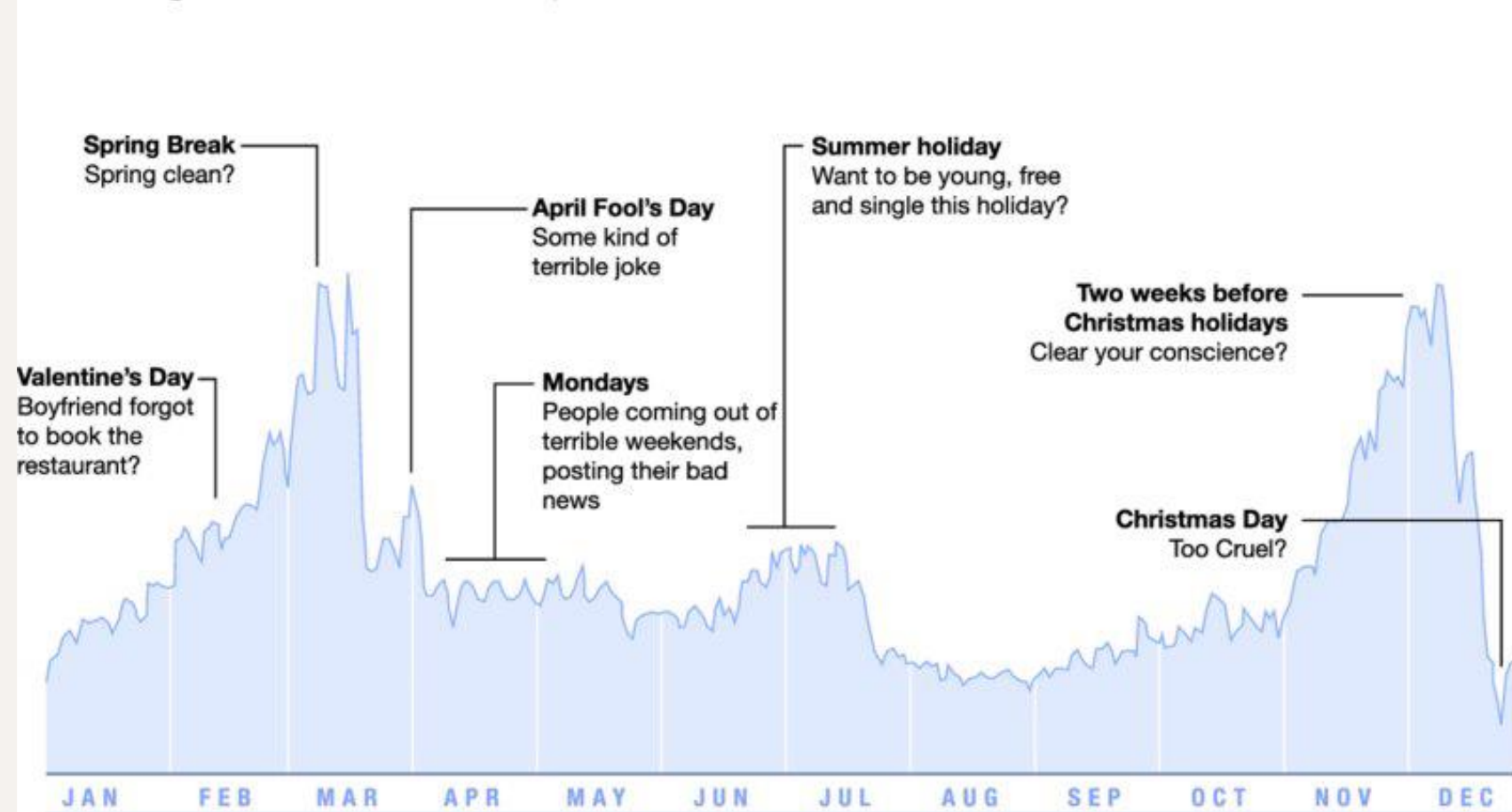
- Text plays a number of roles in communicating with data: use it to **label, introduce, explain, reinforce, highlight, recommend, and tell a story.**
 - Every chart needs a **title** and every axis needs a title (exceptions to this rule will be *extremely* rare).
 - Label explicitly so they can use their brainpower to understand the information, rather than spend it trying to figure out how to read the visual.
 - Leverage preattentive attributes to make those important words stand out.
- 

Text is your friend

A few choice words and phrases make this data so much more quickly accessible than it otherwise would be.

What is missing?

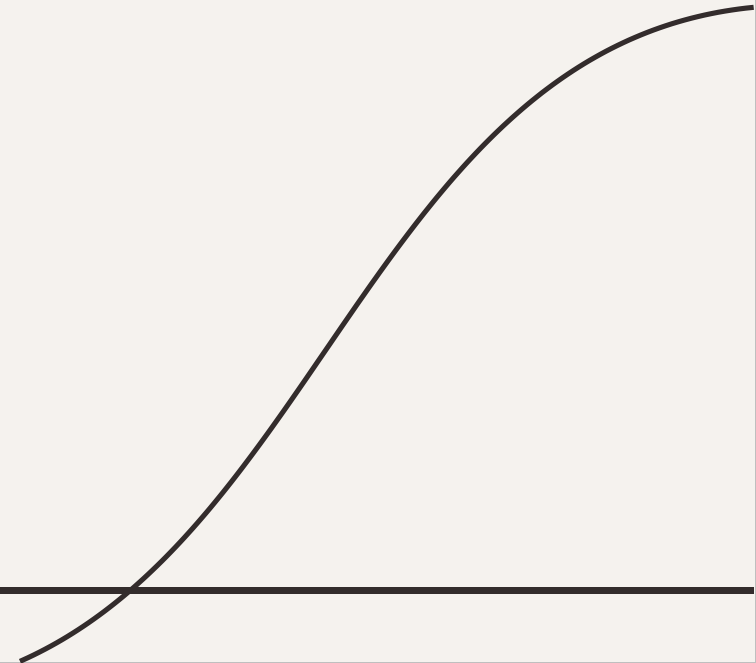
Peak Break-up Times
According to Facebook status updates



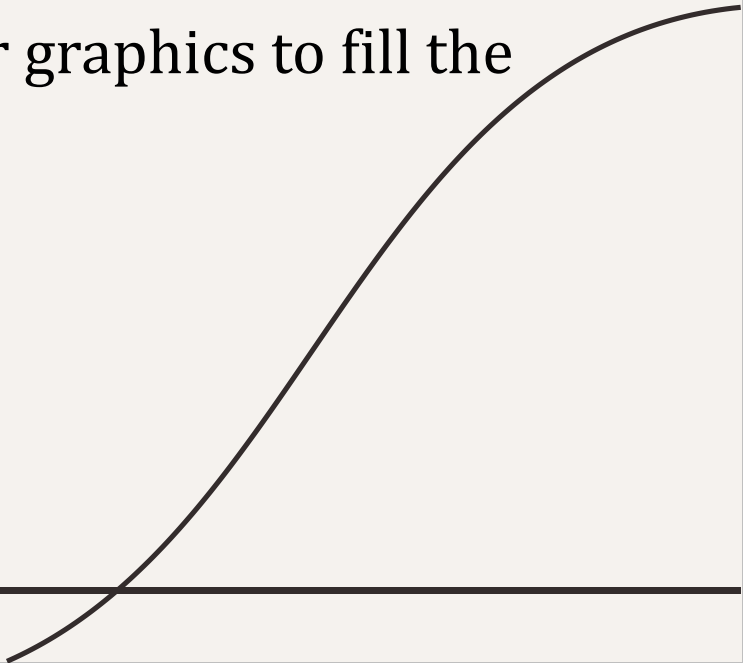
Aesthetics

When it comes to communicating with data, *is it really necessary to “make it pretty?”*

Studies have shown that more aesthetic designs are not only perceived as easier to use, but also more **readily accepted and used over time**, promote creative thinking and problem solving, and foster positive relationships, making people more tolerant of problems with designs.

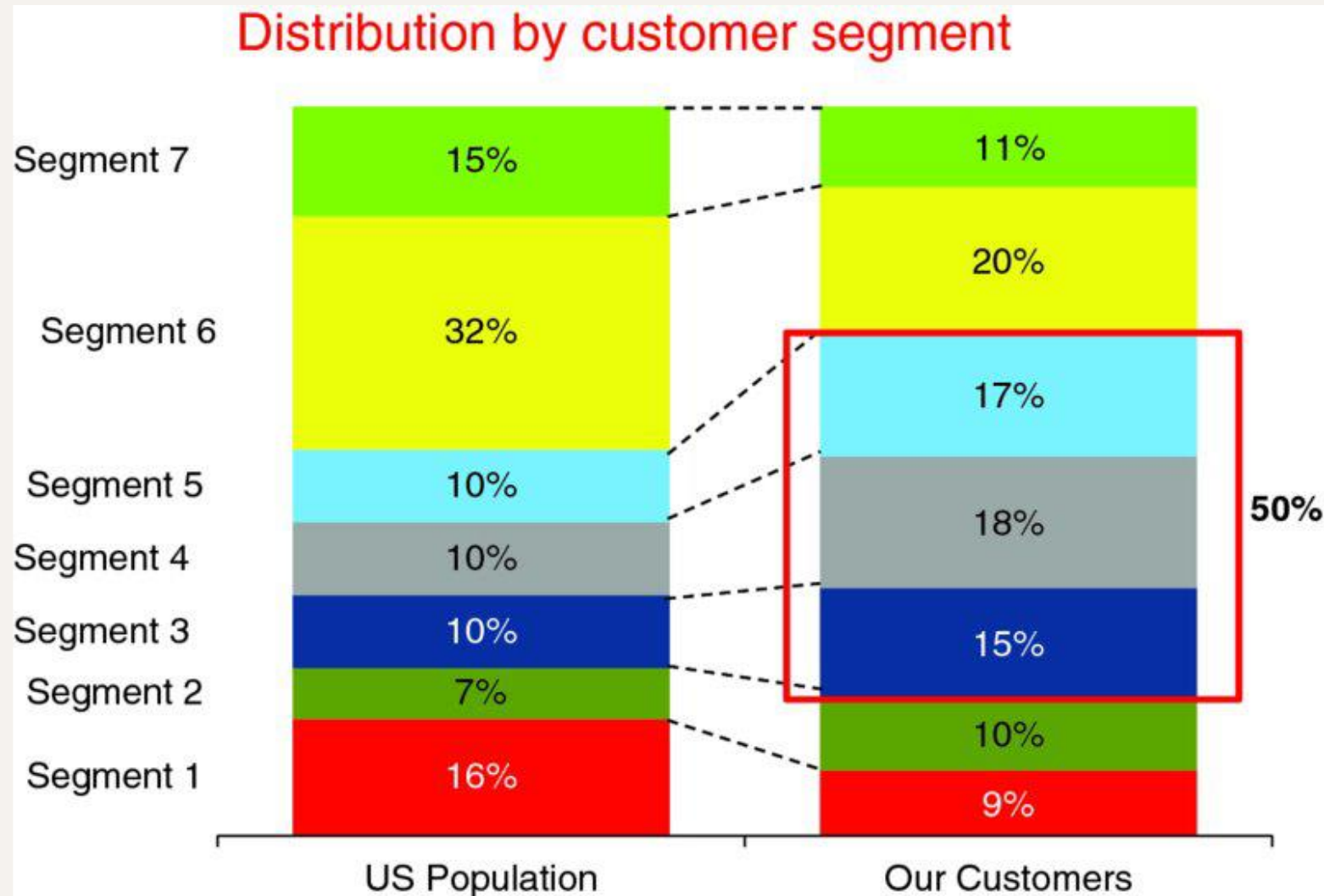


Aesthetics: Best Practices

- 1. Be smart with color.** The use of color should always be an intentional decision; use color sparingly and strategically to highlight the important parts of your visual.
 - 2. Pay attention to alignment.** Organize elements on the page to create clean vertical and horizontal lines to establish a sense of unity and cohesion.
 - 3. Leverage white space.** Preserve margins; don't stretch your graphics to fill the space, or add things simply because you have extra space.
- 

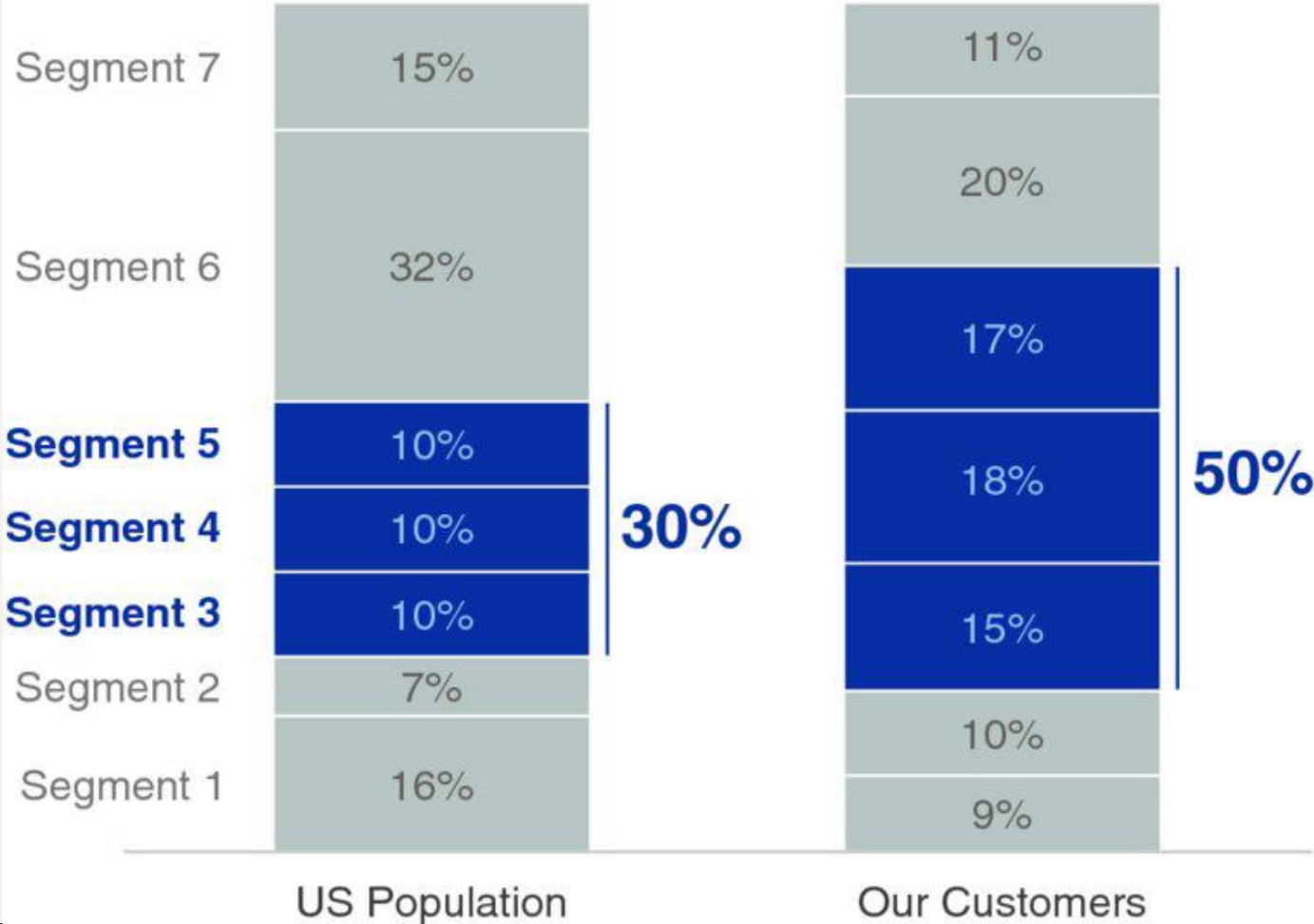
Aesthetics

- Color use?
Overused, too many colors competing for our attention.
- Alignment?
Elements not properly aligned.
- White space use?
Misused, too much of it between the segment titles and data.



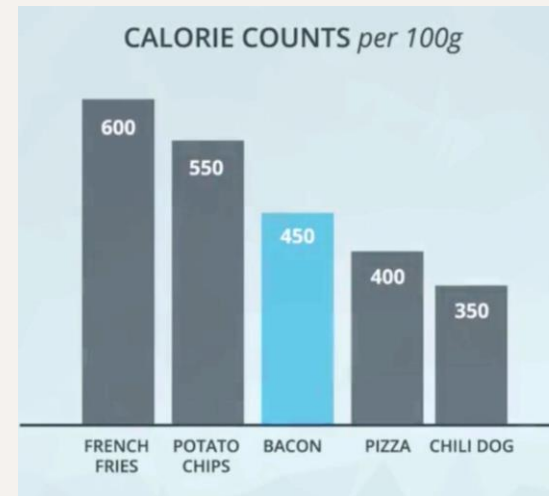
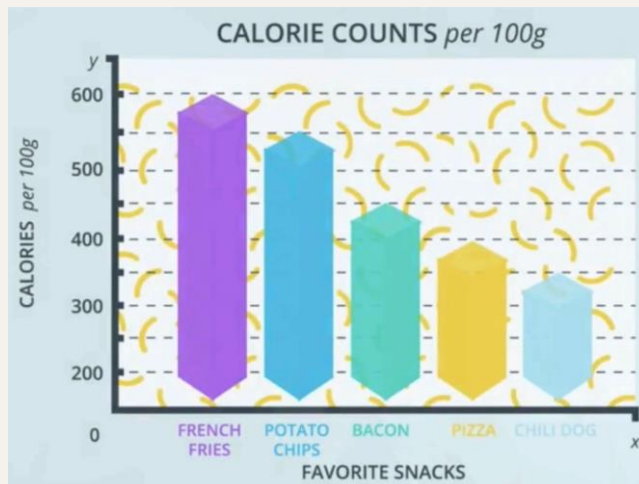
Aesthetics

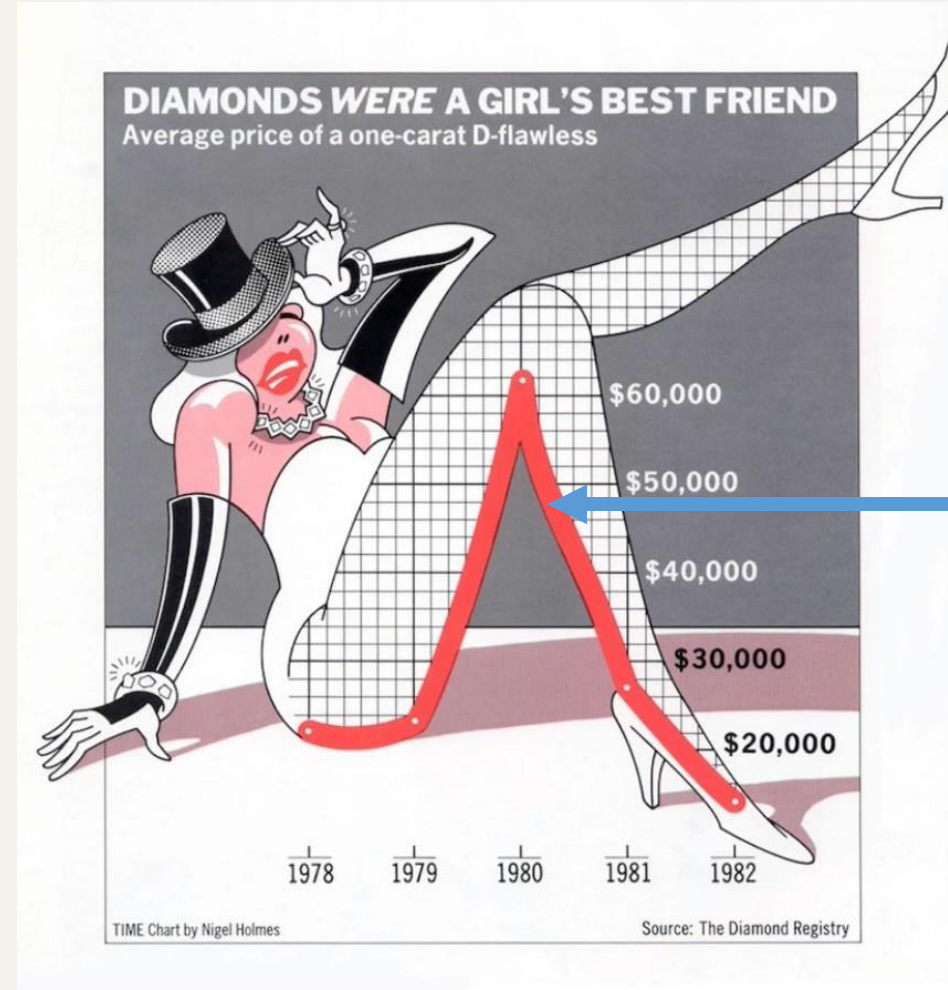
Distribution by customer segment



The Golden Rule

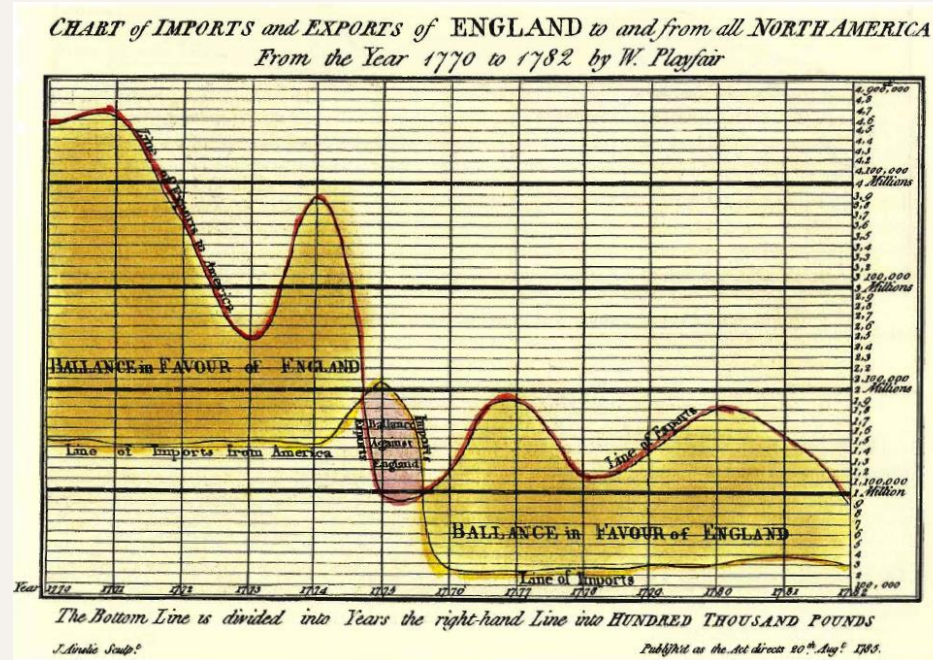
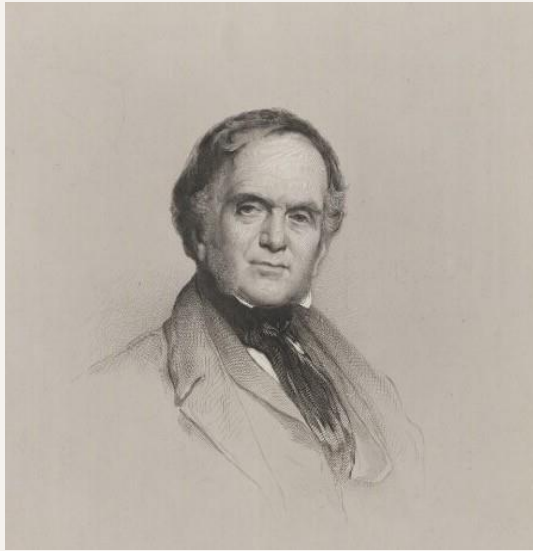
- Maximize the idea clarity while minimizing the following:
 - **Time:** You have the attention of your audience for moments
 - **Ink:** Present more information and less distraction
 - **Space:** Always go for the graph that will give you less space.





Data graphics should draw the user's attention to what matters: the sense and substance of the data, not to something else.

Do you remember him?



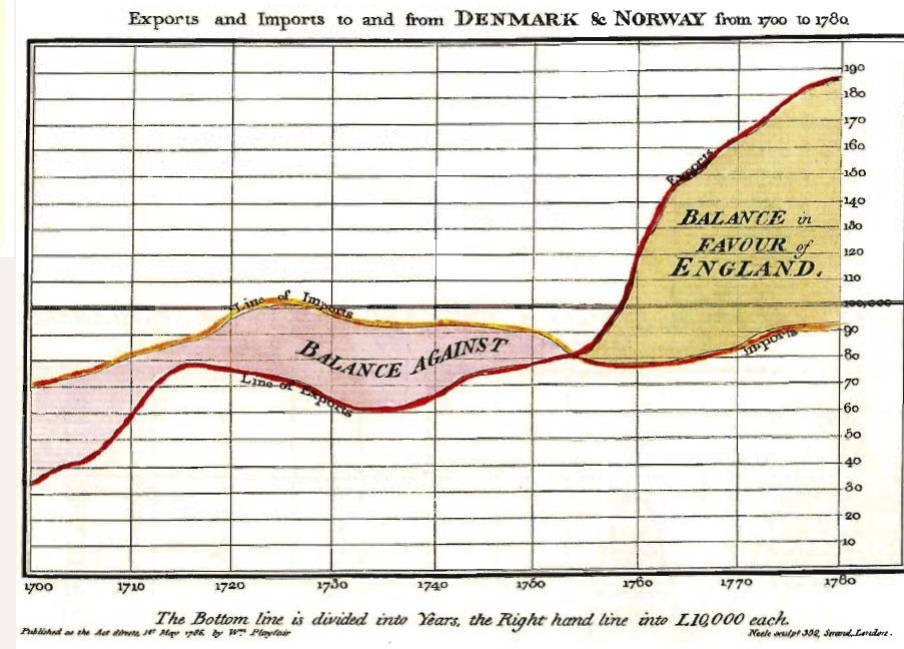
Playfair's very firsts charts devoted too much of their ink to graphical apparatus

William Playfair (1759-1823)

Within a year, Playfair eliminated much of the non-data detail in favour of cleaner design

This improvement illustrates the fundamental principle of good statistical graphics:

Above all else, show the data



Data-ink ratio

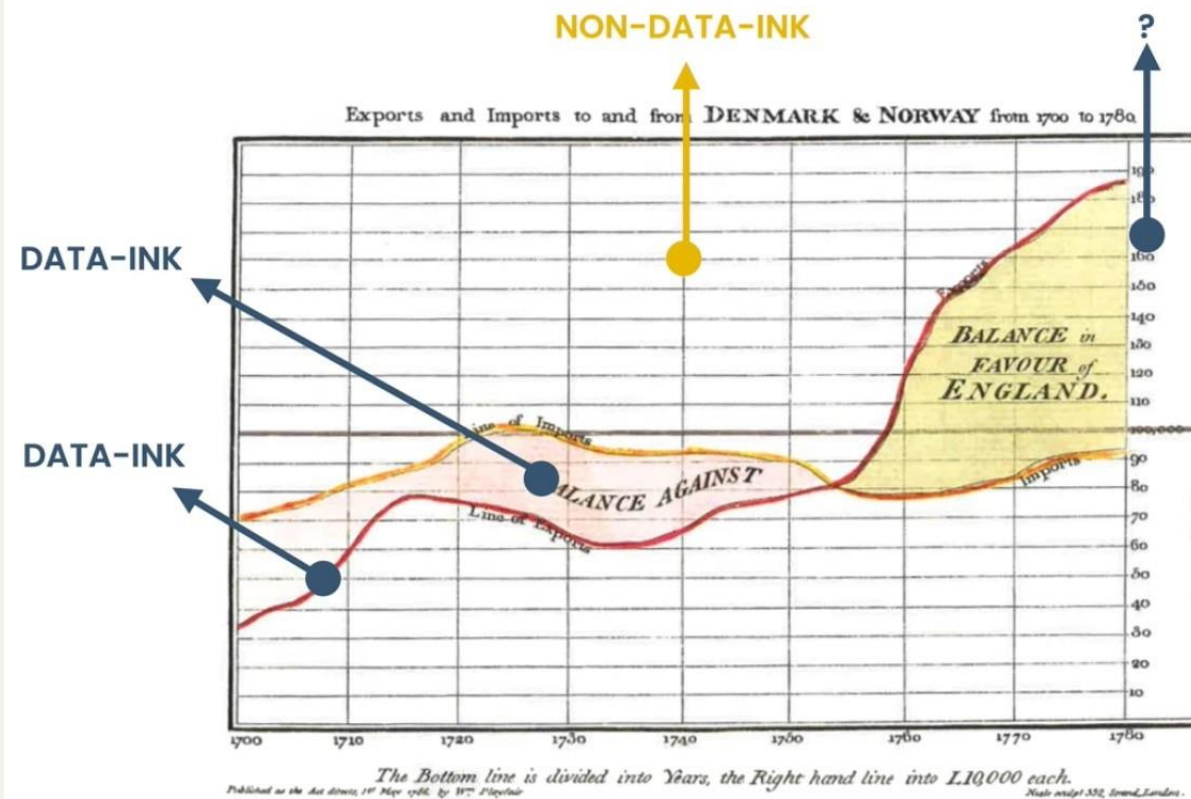
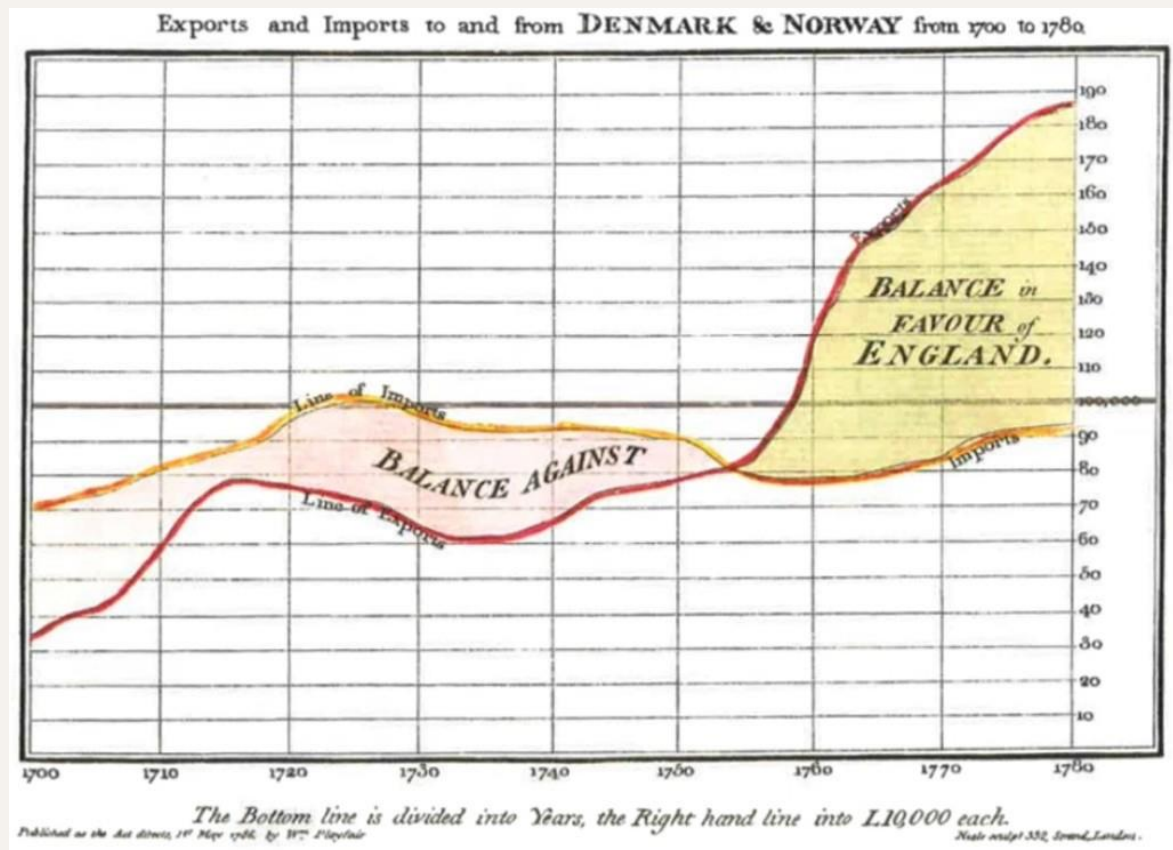
Data graphics should draw the user's attention to what matters.

- **Data-Ink:** The core of the graphic (or non-redundant ink)
- **Non-Data-Ink:** The rest of the ink in the graph

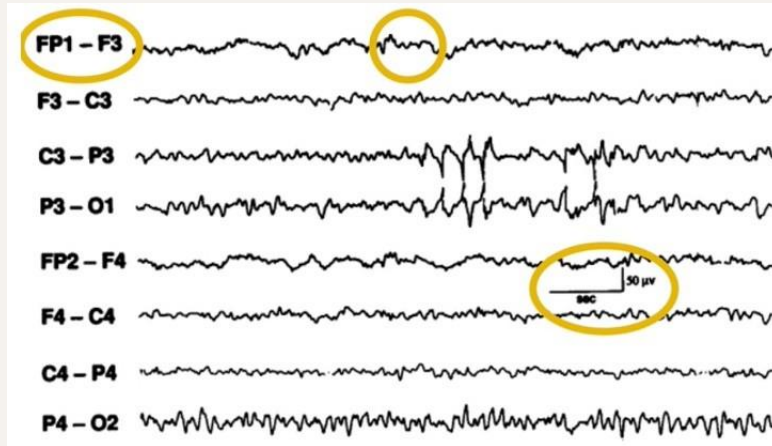
$$\begin{aligned}\text{Data-ink ratio} &= \frac{\text{data-ink}}{\text{total ink used to print the graphic}} \\ &= \text{proportion of a graphic's ink devoted to the non-redundant display of data-information} \\ &= 1.0 - \text{proportion of a graphic that can be erased without loss of data-information.}\end{aligned}$$

Our goal is to make the data-ink ratio as close to 1 as possible (within reason)

Data-ink ratio

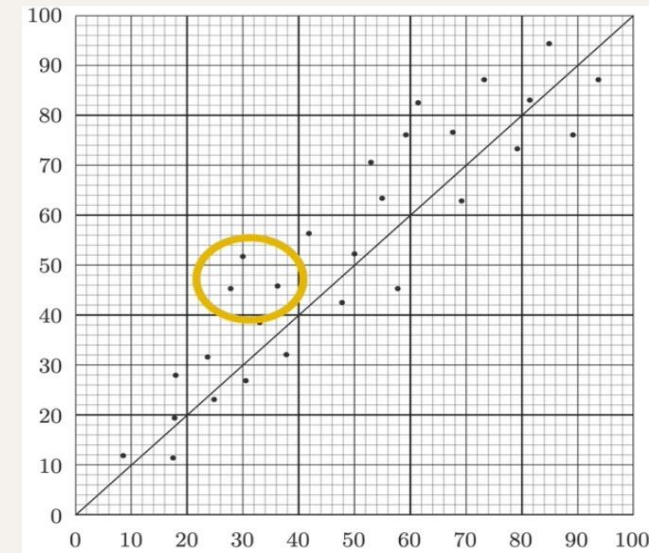


Data-ink ratio: Examples

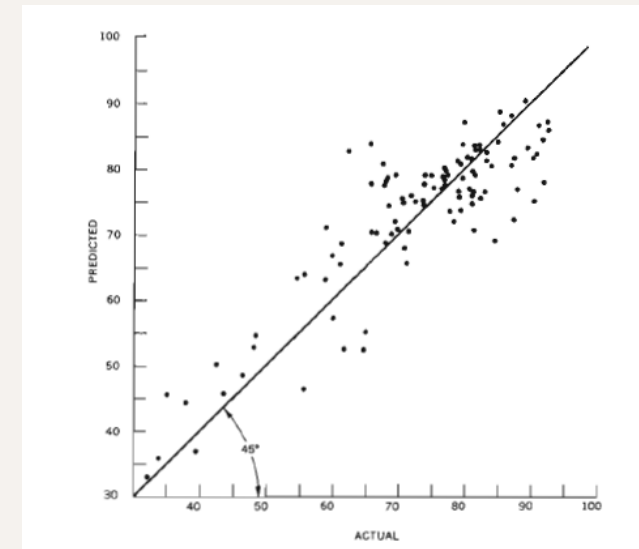


Data-Ink Ratio = Max

Nothing can be erased without losing information in these continuous eight tracks of an electroencephalogram



Data-Ink Ratio = Low



Data-ink ratio

Data-Ink Ratio: Maximizing the share of data-ink

The larger the share of a graphic's ink devoted to data, the better (other relevant matters being equal):

Maximize the data-ink ratio, within reason

Every bit of ink on a graphic requires a reason. And nearly always that reason should be that the ink presents new information

Erase non-data-ink, within reason

Ink that fails to depict statistical information does not have much interest to the viewer of a graphic

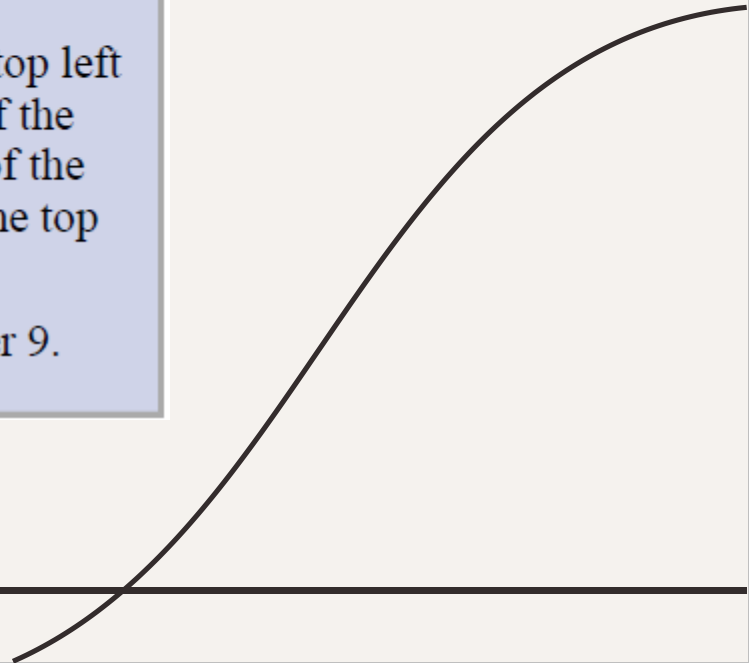
About logical ordering of categories

The logical ordering of categories

When designing any graph showing categorical data, be thoughtful about how your categories are ordered. If there is a natural ordering to your categories, it may make sense to leverage that. For example, if your categories are age groups—0–10 years old, 11–20 years old, and so on—keep the categories in numerical order. If, however, there isn't a natural ordering in your categories that makes sense to leverage, think about what ordering of your data will make the most sense. Being thoughtful here can mean providing a construct for your audience, easing the interpretation process.

Your audience (without other visual cues) will typically look at your visual starting at the top left and zigzagging in “z” shapes. This means they will encounter the top of your graph first. If the biggest category is the most important, think about putting that first and ordering the rest of the categories in decreasing numerical order. Or if the smallest is most important, put that at the top and order by ascending data values.

For a specific example about the logical ordering of data, check out case study 3 in Chapter 9.



Cycle of Visualization

Chart

Visual representation of information

Focuses on a few variables

Dashboard

A collection of charts that presents information about various variables in one location

Audience Driven and Interactive

Data Story

A narrative that uses charts along with text to convey a clear message or conclusion

Author Driven and static

Layout of Dashboard

Understand *where* users look when seeing a dashboard for the first time

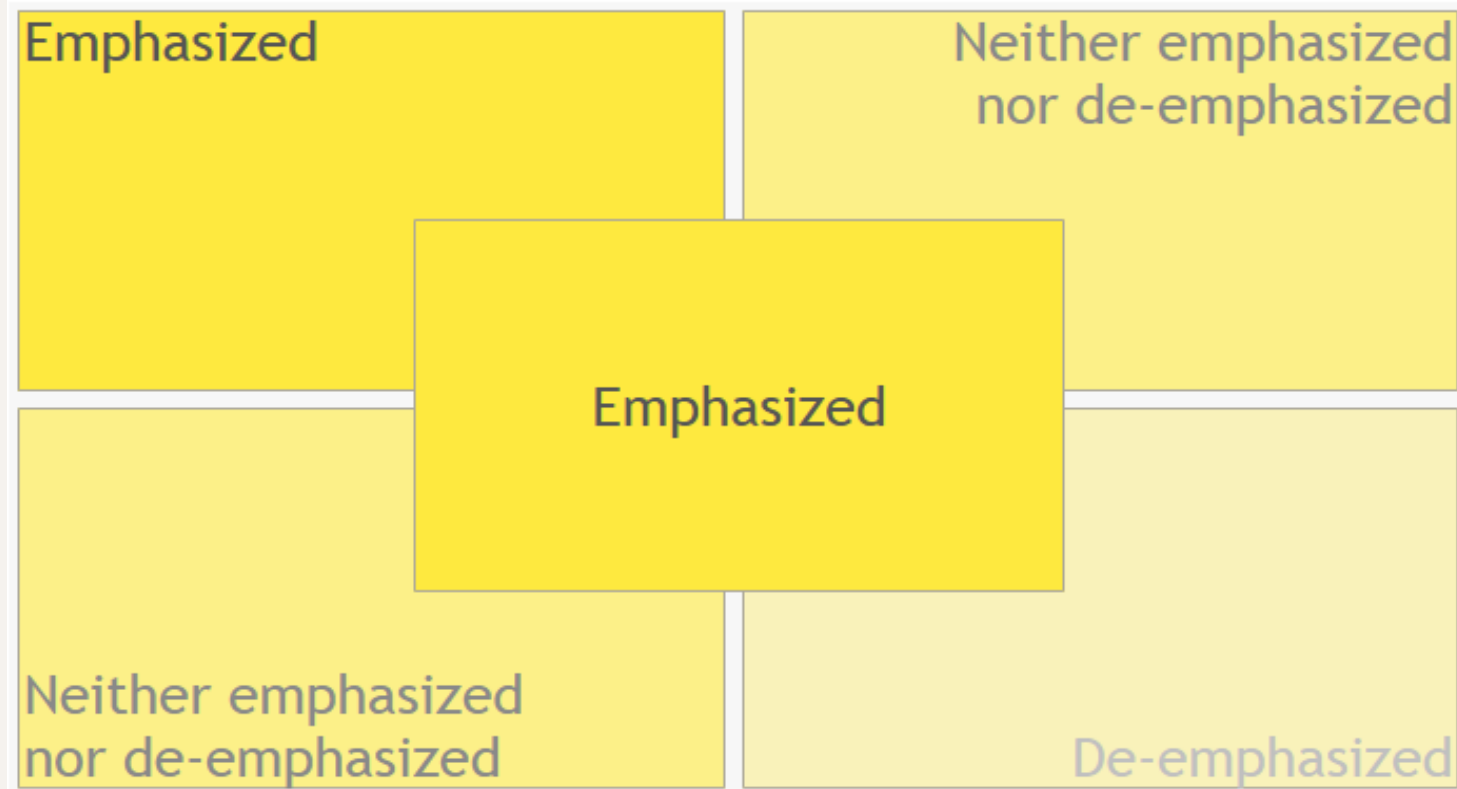
Why Does It Matter?

"I've got 10 seconds..."

- C-level dashboards need fast clarity

"Remember me?"

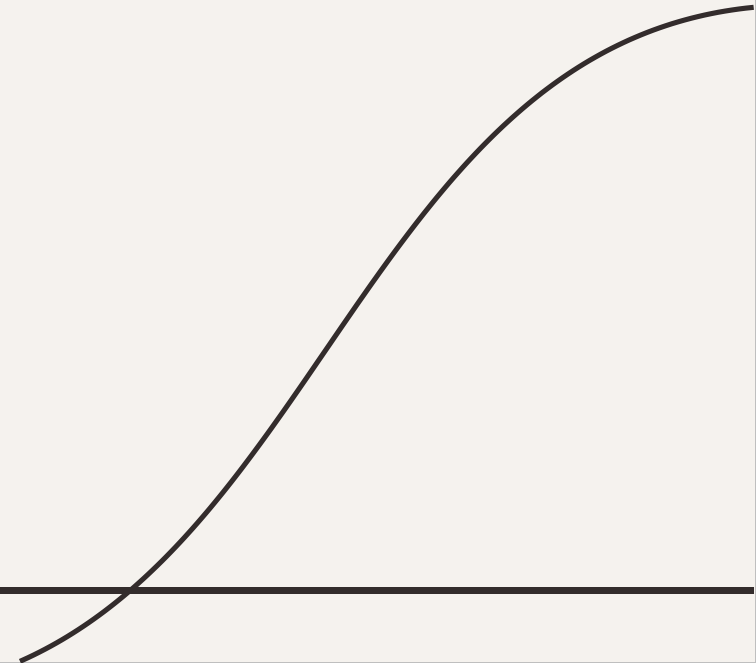
- Dashboards with periodic users need quick reorientation



5 Key Takeaways for Dashboard Designers

1. Big Numbers Get Attention

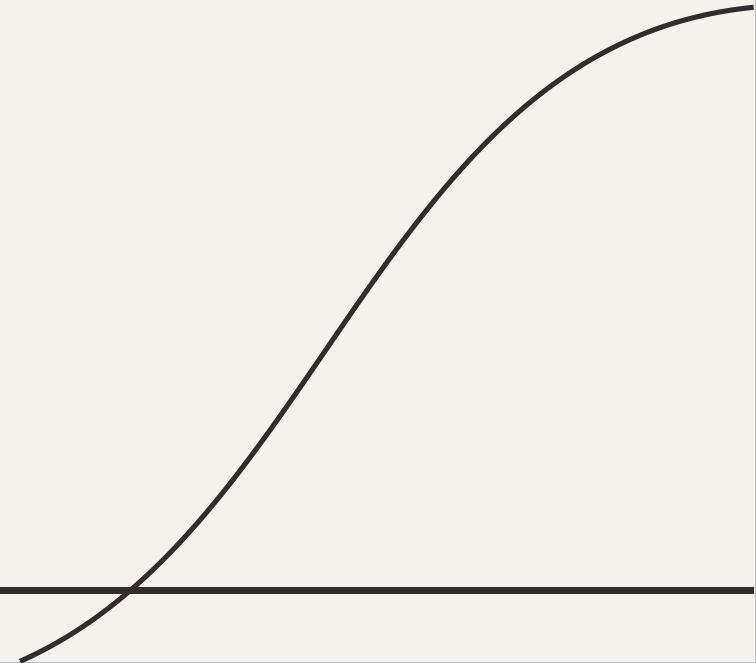
- Large-font numbers = Early and focused attention
- Action: Highlight KPIs or critical metrics using big, bold numbers



5 Key Takeaways for Dashboard Designers

2. Beware of Repetition Fatigue

- Repetitive charts (e.g., multiple KPIs) lose attention as users scan
- Attention highest for:
 - Top-left element
 - First element in a sequence
- **Action: Prioritize order of information**



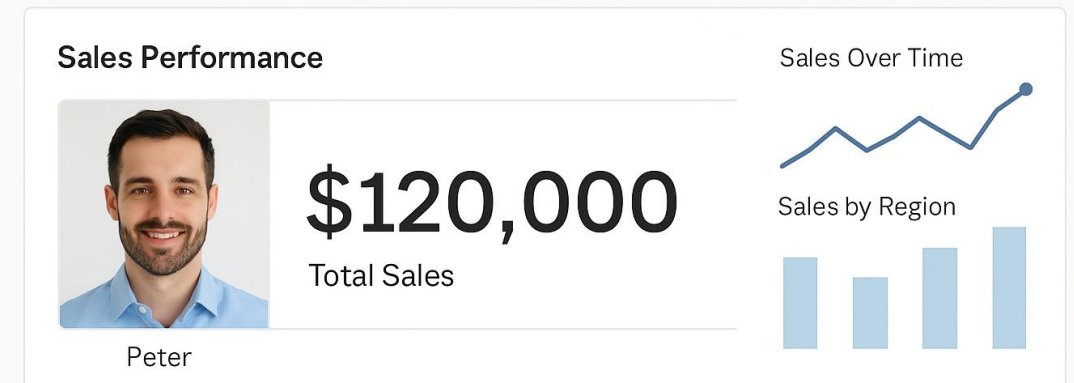
5 Key Takeaways for Dashboard Designers

3. Humans Look at Humans

- Photos or human-like figures *always* attract attention
- Action: Use human imagery *only* if relevant to your message
- Risk: Distracting from data insights

Practical Example: Humans Look at Humans

A photo of the top salesperson was included to “humanize” the data. However, users look at their face first, which could distract from sales figures.



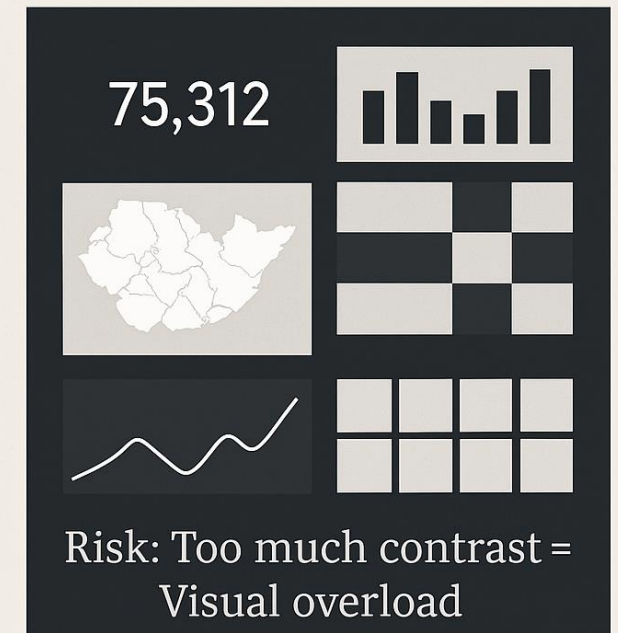
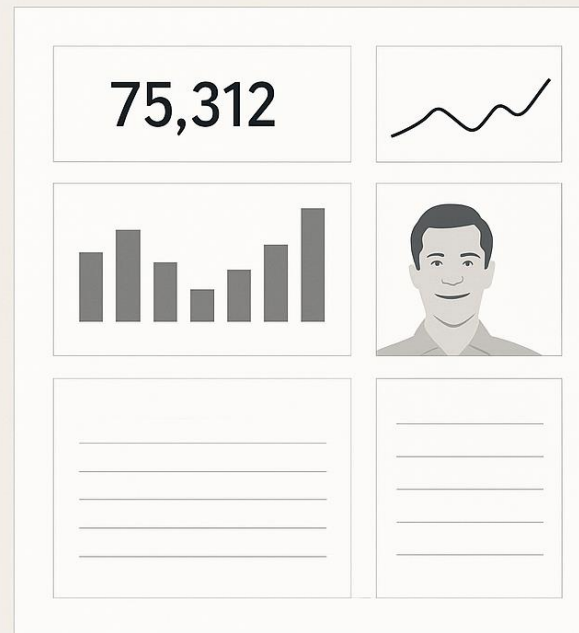
5 Key Takeaways for Dashboard Designers

4. Guide with Contrast

- High contrast areas act as visual guideposts
- Action:
 - Use contrast *sparingly*
 - Create a visual flow or "path" for the eye
- Risk: Too much contrast = Visual overload

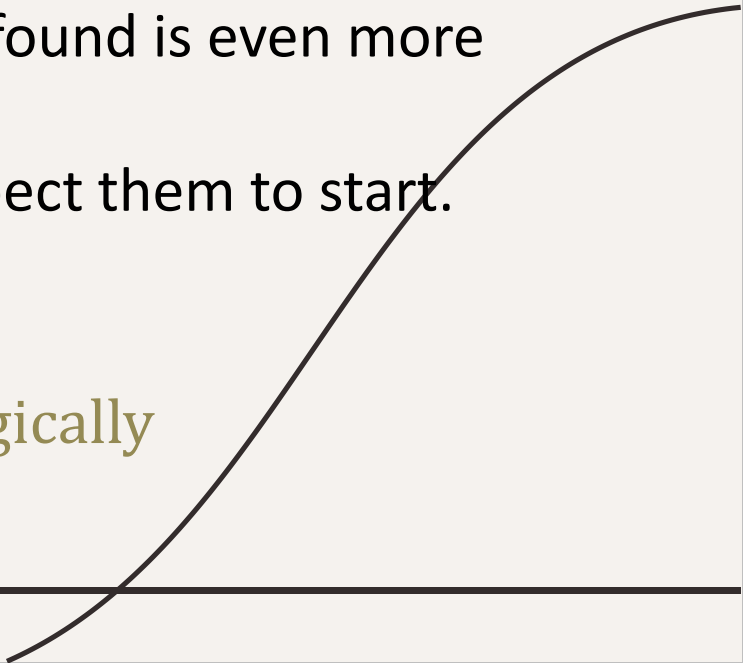
Guide with Contrast

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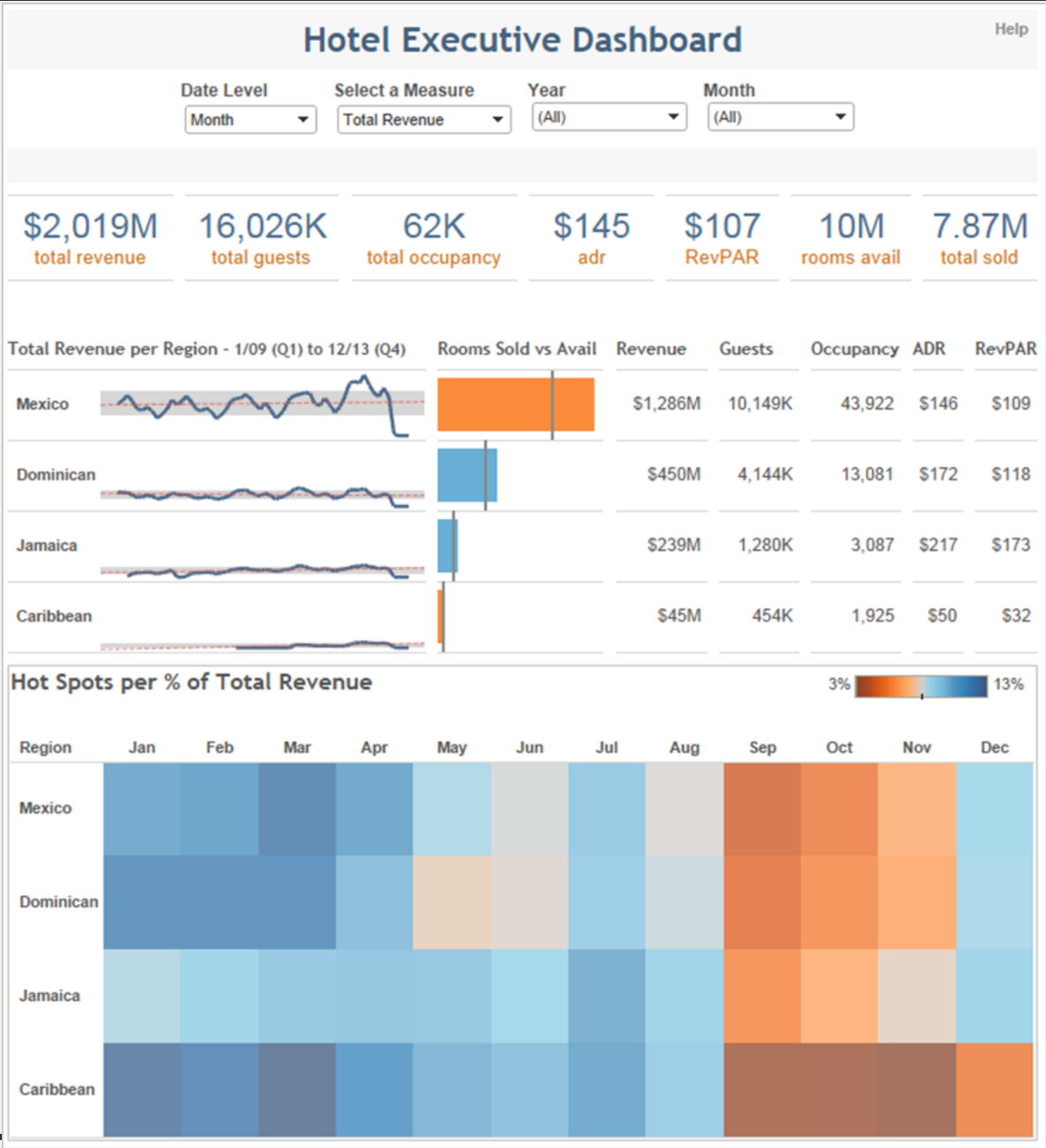
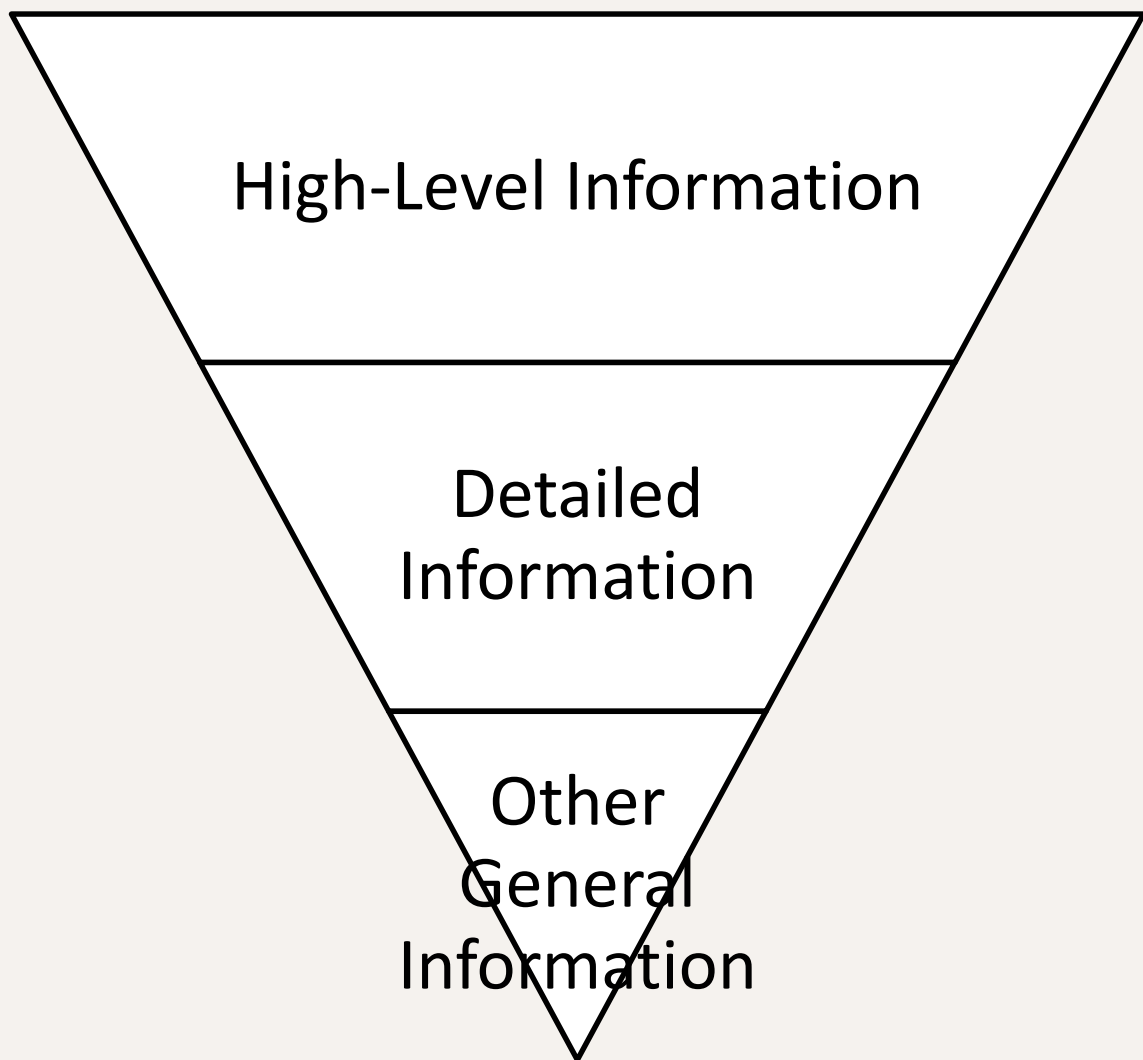


5 Key Takeaways for Dashboard Designers

5. Form Drives Function

- Users follow the dashboard's layout naturally
 - No need to force everything into top-left
 - ***"Put the most important information in the top-left corner because people read left to right."***
 - That's *partially* true — but what the eye-tracking study found is even more powerful:
 - People scan *where the data is* — not just where you expect them to start.
 - **Action:**
 - Be intentional with layout
 - Leverage shapes (grids, columns) to organize data logically
- 

Layout of Dashboard



Thoughts When Creating Dashboards

Objective	Choice of Visualizations	Visual Design	Clutter	Layout
• Goal of Visual (Title or text?) • Defined Audience	• Visuals accurate? • Are visuals related to the objective?	• Fonts and sizes of visuals synchronized? • Colors appropriate? • Context provided? • Axis accurate?	• Ink-Data Ratio • Redudant visuals or text?	• Are their meaningful interactions?